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Considering development equity and landscape patterns to enhance the interaction of nighttime lighting with sustainable development goals

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Nighttime lighting (NTL) remote sensing provides globally consistent and long-term observations of surface nighttime light. It has been widely applied to assess regional economic and social development. However, it remains essential to consider NTL landscape patterns, light equality, and light intensity, to enhance the accuracy and consistency of NTL tools track the key SDGs process, and understand the interaction mechanism between NTL and SDG Score. This research tracks the progress of achieving the 17 SDGs in China, uncovering potential applications of NTL in sustainability science. We found that 84.2% of NTLs in China exhibited an increasing trend, primarily in urban fringe areas, while urban centers showed a declining trend in 2000-2015. The Nighttime Lighting Landscape Index (NTLI), integrating landscape hierarchy, lighting equity, and light intensity, proved optimal for monitoring SDGs, effectively tracking the SDG 1, SDG 8, SDG 9, and SDG 11. By explicitly incorporating landscape patterns and development equity into NTL assessment, this study provides a methodological advance over prior NTL-based approaches and demonstrates how multidimensional lighting information can enhance the robustness and interpretability of sustainability monitoring. These findings build an equity-aware, landscape-informed assessment framework for tracking typical SDGs progress using the NTL data.

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Introduction

In 2015, the United Nations (UN) Summit on Sustainable Development adopted the document “Transforming Our World: The 2030 Agenda for Sustainable Development”. It seeks to promote equitable and coordinated sustainable development at the global scale through the introduction of 17 Sustainable Development Goals (SDGs) (Cf 2015; Gao and Bryan 2017; Schmidt-Traub et al. 2017). As the world’s largest developing country, China has undertaken significant initiatives to support economic-social-environmental sustainable development, including “Poverty Alleviation and Development,” “New Urbanization,” and “Three North Protective Forests”. (Lu et al. 2023; Wang et al. 2023; Zhai et al. 2023). However, uncertainties about the trade-off effects and developments among the SDGs complicate the assurance of resilient and equitable sustainable development (Zhang et al. 2022b; Wu et al. 2022; Cao et al. 2023). China’s subnational regions still experience uneven development, lacking uniform baselines and standards for statistical data; the timing and quantity of sustainability assessment indicators in the annual statistical system vary across regions; and detailed data on the 169 targets of sustainable development is insufficient (Cao 2017; Zhang et al. 2022a). It is imperative to develop a method for assessing sustainable development performance with a consistent baseline over long-time series and large scales. This method can monitor global SDGs performance levels at national, subnational, municipal, and township to identify priority areas for achieving the SDGs (Xu et al. 2020; Warchold et al. 2022).

Q1–Q5

The advancement of remote sensing offers a potent tool for tracking the progress of SDGs across various regions. The UN has explicitly stated that “space is a driver of sustainable development,” and remote sensing as a “spatial tool” is crucial to realizing the SDGs (Guo et al. 2023). Remote sensing facilitates spatial monitoring to elucidate the “economic-social-environmental” development coordination relationship, establishing a unified observational baseline for SDGs globally and providing scientifically credible evidence for decision making (Yu et al. 2023). In addition to Landsat, MODIS, and other resource and environmental satellites supporting the environment dimension of SDGs assessment, Nighttime Lighting (NTL) remote sensing offer a powerful tool for assessing sustainable development in the economic and social dimensions from the perspective of human activities (Prince 2019; Levin et al. 2020; Zheng et al. 2023).

Existing research has clearly demonstrated a high degree of consistency between NTL and socioeconomic-related SDGs, including SDG 1 (Poverty), SDG 7 (Energy), SDG 8 (Economy), SDG 9 (Industry), SDG 10 (Equity), SDG 11 (City and Community) and SDG 12 (Production and Consumption) (Pan and Hu 2018; Guo et al. 2021; Satheendran et al. 2022; Chindarkar and Goyal 2023; Lin et al. 2023; Yamada and Otchia 2025). However, recent studies have increasingly explored the potential of NTL data for monitoring changes in resource and environmental systems. For example, NTL-based indicators have been used to estimate regional carbon emission intensity, thereby supporting climate change mitigation and assessment efforts (Fang et al. 2022; Zhao and Yang 2026). These emerging applications demonstrate the relevance of NTL for assessing environmentally oriented SDGs, particularly SDG 13 (Climate Action). Therefore, it remains unclear which environment-related SDGs can be effectively tracked using NTL, limiting a comprehensive understanding of the applicability of NTL data for monitoring SDGs progress.

In addition, NTL has been widely applied to estimate regional carbon emissions, assess light pollution, measure economic and demographic indicators, and delineate urban areas (Levin et al. 2020; Zhao et al. 2021; Khanduri et al. 2023; Sun et al. 2023). However, most current studies on NTL indices focus on metrics

such as total light intensity (TNLI) and average light intensity (ANLI), overlooking landscape-level features like patch density (PD) and edge density (ED). There is still a need to further consider the spatial equity of regional NTL and to reduce development gaps within regions. This limitation may hinder the ability to capture nuances of inequality and spatial variability in NTL, making it challenging to quantify the equity and disparity of development within regions (Huang and Wang 2019; Yu et al. 2020; Ran et al. 2023). Therefore, it is essential to consider the functional differences between light intensity and landscape patterns in tracking SDGs, and explore the functional efficacy and applicable scenarios of different indicator systems to support sustainable development assessment.

Furthermore, developing assessment methods for sustainable development performance at various scales facilitates the evaluation of NTL instruments’ effectiveness in tracking the SDGs process (Wang et al. 2024b). Although the Global Sustainable Development Report (GSDR), published by the UN Sustainable Development Solutions Network (SDSN), provides an annual sustainable development dashboard for every country, it lacks SDG scores for subnational regions (Sachs et al. 2021). Consequently, several studies have explored strategies for assessing sustainability at subnational levels, such as monitoring spatial-temporal changes in sustainability performance in China subnational area, conducting a city-level sustainability assessment in Shenzhen, China, and evaluating sustainability across 3181 local administrative units in Romania (Xu et al. 2020, 2022; Benedek et al. 2021; Xiao et al. 2023). These studies have gathered extensive SDGs evaluation indicators and endeavored to reconstruct the evaluation system for the 169 targets set by the UN. This approach captures the spatial and temporal changes of the SDG score at subnational levels and below, offering open and transparent information about SDGs progress in each region. These data are used as SDG scores to support the assessment of the specific efficacy of NTL instruments for tracking the SDGs, identifying the best areas for assessment.

Furthermore, there is an interaction mechanism among the SDGs, which numerous studies have substantiated. For instance, the Chinese government’s impact on achieving other SDGs through its precision poverty eradication policy targeting SDG 1 (Poverty), the global quantification of the synergistic, and trade-off effects of SDGs, along with the prediction of potential risks in realizing the SDGs by 2030 (Pradhan et al. 2017; Nash et al. 2020; Issa et al. 2024). It emphasizes that as a monitoring tool for sustainable progress, NTL need quantify its synergistic and trade-off effects with different SDGs to identify priority and focus areas for assessment. The coupled coordination degree (CCD) and the Tapio decoupling model can elucidate the level of coordinated development and interactions among different systems, revealing changes in the coordination trend to determine whether each subsystem’s development is converging (Shi et al. 2020; Han et al. 2022; Wang et al. 2024a). However, the current study has not fully considered the coupling or decoupling level of SDGs with NTL. It is necessary to quantify the impact of NTL and SDG score on the effectiveness of long-term assessments, determine the spatial-temporal changes in the coordination of NTL and SDG score, and underscore the ability of NTL in tracking SDGs process.

Leveraging NTL intensity, lighting equity, and landscape-based indicators, this study aims to develop a comprehensive set of metrics for tracking SDG performance at the subnational level in China. It aims to elucidate the synergistic and trade-off effects between NTL and SDGs, quantify the coupling and decoupling mechanisms, and determine the spatial-temporal variability of the coupling levels. This research intends to address three key

scientific questions: (1) Which SDGs' spatio-temporal progress can be monitored through NTL's light intensity, lighting equity, and landscape patterns? (2) How have NTL and SDGs' synergistic and trade-off effects evolved? (3) What are the spatio-temporal variations in the coupling and decoupling mechanisms between NTL and SDGs in China subnational area. Based on the above scientific questions, this study proposes the following testable hypotheses:

- H1: Integrating nighttime light intensity, lighting equity, and landscape patterns significantly enhances the capacity of NTL indicators to monitor the spatio-temporal progress of SDGs, compared with indicators based on light intensity alone.
- H2: Multidimensional NTL indicators exhibit stronger monitoring capability for SDGs associated with socio-economic development and social equity, while showing weaker or inconsistent associations with environmentally dominated SDGs.
- H3: The synergistic and trade-off relationships between NTL and SDGs vary across space and time, with coupling and decoupling mechanisms exhibiting pronounced subnational heterogeneity in China, reflecting differences in development stage and regional structure.

Research methods

Research area. As the largest developing nation globally by landmass, China contends with persistent sustainable development dilemmas, including regional disparities, the balancing act between economic growth and environment conservation, and pressures on resource provisioning. Furthermore, it must address paramount concerns such as ensuring societal welfare and fostering equitable development. In pursuit of realizing the 2030 SDGs, the Chinese government has initiated targeted programs such as "Precision Poverty Alleviation," "Public Service Enhancement," and "Natural Forest Conservation." These endeavors aim to advance sustainable economic-social-environment coordinated development. To ensure data accessibility and maintain statistical consistency, this study focuses on China's 31 mainland subnational administrative regions, excluding Hong Kong, Macao, and Taiwan (Fig. S1).

Data sources

NTL data. Given that the research time series span the pivotal period of 2013 when the remote sensing sensors of DMSP-OLS and NPP-VIIRS nighttime lights were replaced. To ensure data continuity and mitigate potential calibration errors, this study employs the "nighttime lights similar to NPP-VIIRS from cross-sensor calibration" dataset (15 arcsec, ~500 m) as the primary data source (Chen et al. 2020). This dataset is derived from raw data collected by DMSP-OLS and NPP-VIIRS, employing a convolutional neural network (CNN) with an encoder model to synthesize long-time series data exhibiting exceptional spatial patterns and temporal consistency. It is utilized to quantify NTL intensity, development equity and landscape patterns in 2000–2015 (Fig. S2).

Sustainable development performance. Assessing sustainable development processes is paramount for identifying areas requiring future-oriented interventions. This study employs a widely used SDG score dataset to cover SDGs assessment indicators as possible comprehensively and adequately gauge SDGs progress in China's subnational regions. This dataset accurately computes the SDG score for each of China's subnational level regions for 2000, 2005, 2010, and 2015, ensuring SDG score

comparisons' consistency and scientific validity (Xu et al. 2020). The dataset was constructed with reference to the indicators adopted in the annual Sustainable Development Progress Assessment Reports released by the Sustainable Development Solutions Network (SDSN), and was evaluated using official data sources including the China Statistical Yearbook, China Environmental Statistical Yearbook, China Educational Statistical Yearbook, China Health Statistical Yearbook, China Energy Statistical Yearbook, and China Population Statistical Yearbook. Since the UN did not prioritize the 17 SDGs, equal importance is attributed to each and equal weights are assigned to calculate the subnational SDG score. SDG14 (marine life) is excluded from the average SDG score calculation for inland provinces due to the absence of relevant performance data in some subnational area (Zhang et al. 2022b). To accurately differentiate SDGs in the economic-social-environment dimensions, this study categorized the 17 SDGs into economic SDGs (SDG 1, SDG 2, SDG 8, SDG 9, and SDG 17), social SDGs (SDG 3, SDG 4, SDG 5, SDG 10, and SDG 16), and environmental SDGs (SDG 6, SDG 7, SDG 11, SDG 12, SDG 13, SDG 14, and SDG 15) (Yin et al. 2023). To avoid overestimating NTL's tracking of SDG progress, we selected highly relevant SDGs for further analysis, including SDG 1, SDG 7, SDG 8, SDG 9, SDG 10, SDG 11, SDG 12, and SDG 13 (Table S1).

Research conceptual framework. This study employed NTL data and sustainable development indicators as the primary datasets. Based on NTL observations, we first examined the long-term trends of night-time light dynamics and quantified three landscape pattern and intensity indices, namely the Nighttime Lighting Intensity Index (NTI), Landscape Pattern Index (LI), and Nighttime Lighting Landscape Index (NTLI). Subsequently, we investigated the trade-offs and synergies between NTL and typical SDGs, as well as their coupling coordination degree based on the spearman correlation coefficient, coupling coordination degree model, and Tapio model. Furthermore, we explored the spatial hotspots and temporal variability of the coupling relationship based on the kernel density estimation method and spatial hotspot analyze, thereby providing empirical evidence for the use of NTL approach to track regional sustainable development. The overall research framework is illustrated in Fig. 1.

Night lighting data processing

Trend analysis of light intensity changes. Sen trend analysis and the Mann-Kendall test were used to obtain the trend of NTL and its statistical significance from 2000 to 2015. Sen trend analysis and the Mann-Kendall test are widely used in trend analysis of time series. They compensate for the shortcomings of the traditional linear method, which is susceptible to the interference of outlier data or large measurement error, and are suitable for the trend analysis of long time series data (Sun et al. 2022; Zhou et al. 2023). The results of MK-Sen trend analysis are judged using " $|Z|$ ", according the " $|Z|$ ", we recognized the eight trend levels of NTL.

Since raster data typically exhibit spatial autocorrelation, directly applying the MK-Sen test may lead to inflated significance levels. To address this issue, Moran's I was first calculated as a diagnostic measure of spatial dependence. Based on the Moran's I value from 2000 to 2015; the effective sample size (Neff) method was then employed to adjust the MK-Sen test results. However, the strong spatial autocorrelation caused an excessive correction, which obscured the temporal trends of nighttime light variations across most regions. To mitigate this overcorrection and better capture the spatio-temporal dynamics of nighttime light intensity, a correction factor (α) was introduced

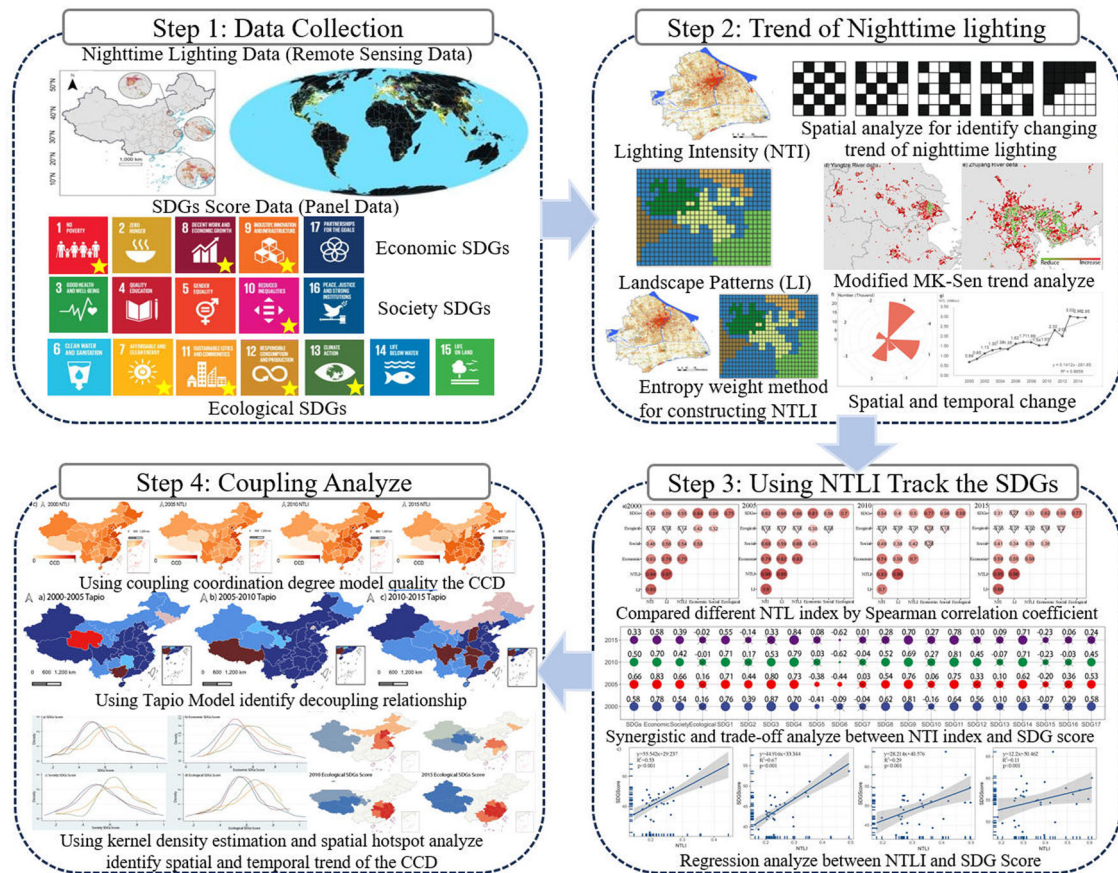


Fig. 1 The research conceptual framework.

Table 1 Nine levels defining the temporal variation of NTL intensity.

β	Z	Trend type	Trend features
$\beta > 0.005$	$ Z > 2.58$	4	Extremely significant increase
	$1.96 < Z \leq 2.58$	3	Significant increase
	$1.65 < Z \leq 1.96$	2	Slightly significant increase
	$ Z \leq 1.65$	1	No significant increase
$-0.005 < \beta < 0.005$	Z	0	No change
$\beta < -0.005$	$ Z \leq 1.65$	-1	No significant reduction
	$1.65 < Z \leq 1.96$	-2	Slightly significant reduction
	$1.96 < Z \leq 2.58$	-3	Significant reduction
	$ Z > 2.58$	-4	Extremely significant reduction

(Gutiérrez-Hernández and García 2024).

$$N_{eff} = \frac{N}{1 + 2 \sum_{k=1}^{N-1} \rho_k (1 - \frac{k}{N})} \quad (1)$$

$$N_{eff}^* = N * \left[\frac{1}{1 + (N-1)\rho} \right]^\alpha \quad (2)$$

Among them, N mean the origin sample, ρ_k mean the Moran's index, stand for the spatial correlation, while $N_{eff} < N$, the significance value (Z) will down. Considering the Moran's index have more influence with formula, refer to the older research, we add the artificial scaling factor α to control the scaling intensity. We set the scaling factor $\alpha=0.1$ to consider the spatial correlation and raster scale change trend at the same time.

We also revised variance (Formula 3) and Z -value calculation (Formula 4) based on the N_{eff}^* , apply the level system to define the temporal variation of NTL intensity. Based on the Z -value and

trend, the change trend is categorized into nine levels (Table 1)

$$Var^*(S) = \frac{N_{eff}^* (N_{eff}^* - 1) (2N_{eff}^* + 5)}{18} \quad (3)$$

$$Z^* = \begin{cases} \frac{S-1}{\sqrt{Var^*(S)}} > 0 \\ 0 & S = 0 \\ \frac{S+1}{\sqrt{Var^*(S)}} < 0 \end{cases} \quad (4)$$

Constructing the NTL index. By integrating the spatial spillover and agglomeration effects crucial for assessing China's SDGs, and drawing upon the emphasis on light intensity and landscape pattern indicators in previous studies. This study amalgamates a comprehensive set of indicator systems for NTL landscape patterns and light equity (Table 2).

Table 2 Indicators of the NTLI system and their weights, including landscape pattern and light intensity.

Pattern	Indicator	Describe	Weight
Landscape Pattern	PLAND	Percentage of Landscaped Area	0.08
	ED	Edge Density	0.07
	PD	Patch Density	0.06
	LSI	Landscape Shape Index	0.25
	CLUMPY	Clumpy Index	0.11
	COHESION	Patch Cohesion Index	0.02
Intensity Pattern	TNTL	Total Lighting Intensity	0.03
	LAREA	Total Lighting Illuminated Area	0.16
	SANLI	Total Intensity/Lighting Area	0.05
	ANLI	Total light intensity/ Total area of administrative area	0.09
	STD	Standard deviation, all areas	0.03
	PI	Poverty, lower 10% total nighttime light/ Total intensity	0.03
	AI	Affluence, upper 10% total nighttime light/ Total intensity	0.02

Note: The weight values in the table are rounded, so the weights do not sum to 1. However, the NTLI is calculated according to the original weight values.

The landscape pattern level indicator system comprises area indicators, edge indicators, shape indicators, and physical connectivity indicators. Area indicators encompass the percentage of landscape area (PLAND) and patch density (PD), delineating the spatial dominance of NTL. Edge indicators focus solely on edge density (ED), elucidating the continuity of NTL's landscape. Shape indicators incorporate the landscape shape index (LSI), reflecting the complexity of NTL's landscape. Physical connectivity indicators include the patch cohesion index (COHESION) and clumpy index (CLUMPY), illustrating the spatial aggregation status of NTL (Huang and Wang 2019; Yu et al. 2020; Han et al. 2022; Chen et al. 2023; Ran et al. 2023).

The light intensity indicators encompass three facets: light intensity, light range, and light equity. Light intensity indicators comprise total nighttime lighting intensity (TNTL), average nighttime lighting index (ANLI), and total intensity lighting area (SANLI). The light range metrics include the total lighting illuminated area (LAREA), which is used to describe the range of NTL in an area. Light equity metrics including standard deviation (STD), poverty index (PI), and abundance index (AI) are used to characterize the NTL equity within a region. In calculations, ignored all pixels with value 0, only considered the STD, PI and AI for pixels where illumination was present.

Formulated the NTLI utilizing the entropy weighting method. To assess the effectiveness of NTL intensity and landscape pattern in monitoring SDG score, separately quantified the NTI and LI while computing the NTLI. This facilitated a comparison of the efficacy among the three NTL indices, NTLI, NTI, and LI, in tracking SDG score. The final weights were determined as average across 2000, 2005, 2010, and 2015 (Table 2). The entropy weighting method assigns weights based on the information content of each variable, thereby maximizing the use of variability in the original data. To evaluate the robustness of the results, outcomes derived from the entropy-weighted approach were compared with those obtained using an equal-weighting scheme (Table S7 and Fig. S16). The indicators per unit of light intensity carry greater weights than indicators reflecting the cumulative light intensities, underscoring the principle of averaging rather than totaling. To ensure that the NTLI, NTI, LI and SDG Score have the same scale, we multiplied the NTLI, NTI, LI results by 100 so that both the NTLI, NTI, LI and SDG Score range from 0 to 100. For more details on the constructing the NTL index, please refer to Supplementary Method.

Coupling of the NTL index change with the SDGs process

Identifying synergistic and trade-off effects between NTLI and SDG Score. The Spearman correlation coefficient was employed to

evaluate the trade-off and synergistic impacts between NTL and typical SDGs (SDG 1, SDG 7, SDG 8, SDG 9, SDG 10, SDG 11, SDG 12, and SDG 13). It is widely utilized to discern overarching relationships beyond linear correlations between two variables (Pearson's 2011). Unlike Pearson's correlation coefficient, Spearman's correlation captures nonlinear associations between variables, does not necessitate normal data distribution, and mitigates the influence of outliers (Zhang et al. 2022b, 2024). Spearman's correlation coefficient values surpassing 0.6 signify a synergistic effect between the two systems (a positive correlation, implying advancement in one goal facilitates advancement in the other). In comparison, values below -0.6 denote a trade-off effect between the two indices (negative correlation, suggesting progress in one goal impedes progress in the other) (Pearson's 2011; Wei et al. 2023).

Coupled Coordination Degree Model (CCDM). The Coupled Coordination Degree Model (CCDM) has been widely used to compare relationships between systems and to help understand them (Shi et al. 2020; Yin et al. 2023). Coupling measures the degree of interaction between systems, and the coupled coordination degree (CCD) measures the level of coupling of an integrated system based on coupling (Xiao et al. 2022; Tao et al. 2023). The CCD was calculated to obtain the coupling coordination relationship between NTL Index and SDG score. Despite its effectiveness in revealing overall coupling patterns, CCDM does not explicitly represent causal relationships and need sensitive to indicator selection and weighting schemes. We employed the CCDM to provide an intuitive and comparable measure of the coupling relationship between NTLI and SDG scores across regions and periods, making it suitable for large-scale spatial and temporal assessment of system-level interactions.

Tapio decoupling modeling. Since NTL index may present a decoupling effect with SDGs performance as a constant change in the type of coupling on a long time series scale. Using Tapio model to quantify the decoupling mechanism of NTL and SDGs (Tapio 2005), revealing the decoupling mechanism can help to further grasp the synergistic effect of the change of the NTL index system with the performance of SDGs. Nevertheless, the Tapio model is descriptive in nature and sensitive to short-term fluctuations, and it does not account for time lags or underlying driving mechanisms. In this study, it is applied as a complementary tool to CCDM, enabling a dynamic interpretation of the temporal evolution of coordination and decoupling between

Table 3 Tapio decoupling grading of NTL index and SDG score.				
Decoupling category	Decoupling status	ΔNTLI	ΔSDG score	α _n
Negative decoupling (ND)	Strong negative decoupling (SND)	> 0	< 0	< 0
	Weak negative decoupling (WND)	< 0	< 0	0-0.8
Coupling (C)	Expansive negative decoupling (END)	> 0	> 0	> 1.2
	Recessive coupling (RC)	< 0	< 0	0.8-1.2
	Expansive coupling (EC)	> 0	> 0	0.8-1.2
Decoupling (D)	Recessive decoupling (RD)	< 0	< 0	> 1.2
	Weak decoupling (WD)	> 0	> 0	0-0.8
	Strong decoupling (SD)	< 0	> 0	< 0

NTL and SDGs:

$$\alpha_n = \frac{\frac{\Delta NTLI}{NTLI_0}}{\frac{\Delta SDG}{SDG_0}} \tag{5}$$

α_n denotes the decoupling index, ΔNTLI,ΔSDG denote the change of NTL index and SDG score from the base year. ΔNTLI₀ and ΔSDG₀ denote the NTL index and SDG score in the base year. The decoupling relationship between NTL index and SDG score is categorized into three types of decoupling states: negative decoupling, coupling, and decoupling, which contains 8 states in total (Table 3).

Revealing temporal changes in CCDs

Temporal variation of CCDs. Kernel density estimation (KDE), as a nonparametric method with good statistical properties, is used to describe the distribution pattern of random variables, and continuous curves are utilized to show the CCD temporal variation differences and dynamic evolutions from the perspective of data (Katkovnik and Shmulevich 2002; Silverman 2018). By examining shifts in the location, shape, and dispersion of KDE curves across 2000-2015, this analysis captures whether the coupling relationship between NTL and SDGs exhibits convergence, polarization, or persistent inequality.

Extracting hotspots of CCD spatial variation. Hotspot analysis was employed to identify noteworthy regions where CCDs exhibit spatial concentration, thereby delineating the spatial hotspot regions of CCDs at the subnational level in China 2000-2015. By calculating the hot and cold spots of CCDs, the spatial clustering effect and spatial variability of CCDs can be effectively discerned, furnishing spatial reference evidence for the NTL to assess the reliability of the sustainable development process. Identifying CCD hotspots and coldspots facilitates the interpretation of spatial inequalities in sustainable development processes. It can help reveal regional patterns of coupling and decoupling. These spatial patterns indicate where development coordination is consistently strong and where targeted interventions may be required to address long-term structural disparities.

For detailed information on the methodology section and the calculation process, please refer to Supplementary Methods S1-S5. Moreover, to ensure the robustness of the results, we designed a sensitivity analysis. For the specific methodology, please refer to Supplementary Methods S6.

Using the nighttime lighting indices track the sustainable development goals

Spatio-temporal patterns change trends of NTL in China. From 2000 to 2015, China’s average NTL intensity increased rapidly, with an average annual growth rate of 22.26%. (*p* < 0.001), soaring from 0.05 to 0.22. Concurrently, the SDG score experienced a notable surge, boasting an average annual growth rate of 1.46% (*p* < 0.01), with the average for China’s subnational level regions

escalating from 45.47 to 55.44. However, the pace of SDG score advancement notably needs to catch up to the burgeoning NTL intensity, suggesting a potential developmental imbalance between the two. Over the same period, a staggering 98.31% of China’s regions witnessed no alteration in NTL intensity, while 1.58% exhibited an increasing trend and 0.11% showed a declining trend. When disregarding regions devoid of NTL, 20.49%, 38.88%, 12.41%, and 21.89% of China’s territories manifested extremely significant, significant, slightly significant, and no significant escalations in NTL intensity, respectively. Besides, 0.17%, 0.87%, 0.69%, and 4.61% of China’s areas displayed extremely significant, significant, slightly significant, and no significant declines in NTL intensity, respectively (Fig. 2).

Descriptive statistics of NTL and SDG score indices. Table 4 presents the descriptive statistics of NTL and SDG score indices. Among them, the SDG score, Ecological SDG score, and Society SDG score exhibited the most rapid growth between 2010 and 2015, whereas the Economic SDG score experienced a sharp increase between 2005 and 2010. The Nighttime Lighting Landscape Index (NTLI) maintained a stable upward trend, the Nighttime Lighting Intensity Index (NTI) grew rapidly between 2010 and 2015, while the Landscape Pattern Index (LI) showed stagnation in growth during the same period.

Efficacy of different NTL indices to assess SDG score. NTL, LI, and NTLI exhibit no correlation with environmental SDG score, implying NTL’s limited capacity to reflect shifts in environmental SDGs, with the assessment scope primarily focusing on economic and social SDGs. As of 2015, LI is no longer significantly associated with the SDG score. In 2000 and 2010, NTI displayed considerably higher correlation coefficients with NTLI and SDG score; however, by 2005 and 2015, NTLI surpassed NTI in correlation with SDG score. Concerning the relationship of NTLI, LI, and NTI with SDG score in the economic and social dimensions, NTLI exhibits the most consistent correlation coefficient. In contrast, the correlation coefficient between NTLI and LI, which represents the nighttime light landscape pattern, showed a more significant difference. Integrating the LI indicator into the NTI metric would bolster the correlation between NTLI and SDG score. By considering the inclusion of landscape pattern indicators alongside traditional light intensity metrics, the efficacy of NTLI in tracking SDG score would be enhanced. Consequently, NTLI demonstrates heightened sensitivity and stability in observing SDG score trends (Fig. 3a).

From 2000 to 2015, China’s subnational SDG scores, economic SDG scores, and environmental SDG scores all exhibited a fluctuating upward trend. However, environmental SDG scores specifically rose from 2000 to 2005 and 2010 to 2015, while experiencing a decline from 2005 to 2010. NTI, LI, and NTLI all exhibit an increasing trend among the three sets of NTL indices, yet both LI and NTLI show significant growth between 2005 and 2010. In contrast, NTI only experiences notable growth between

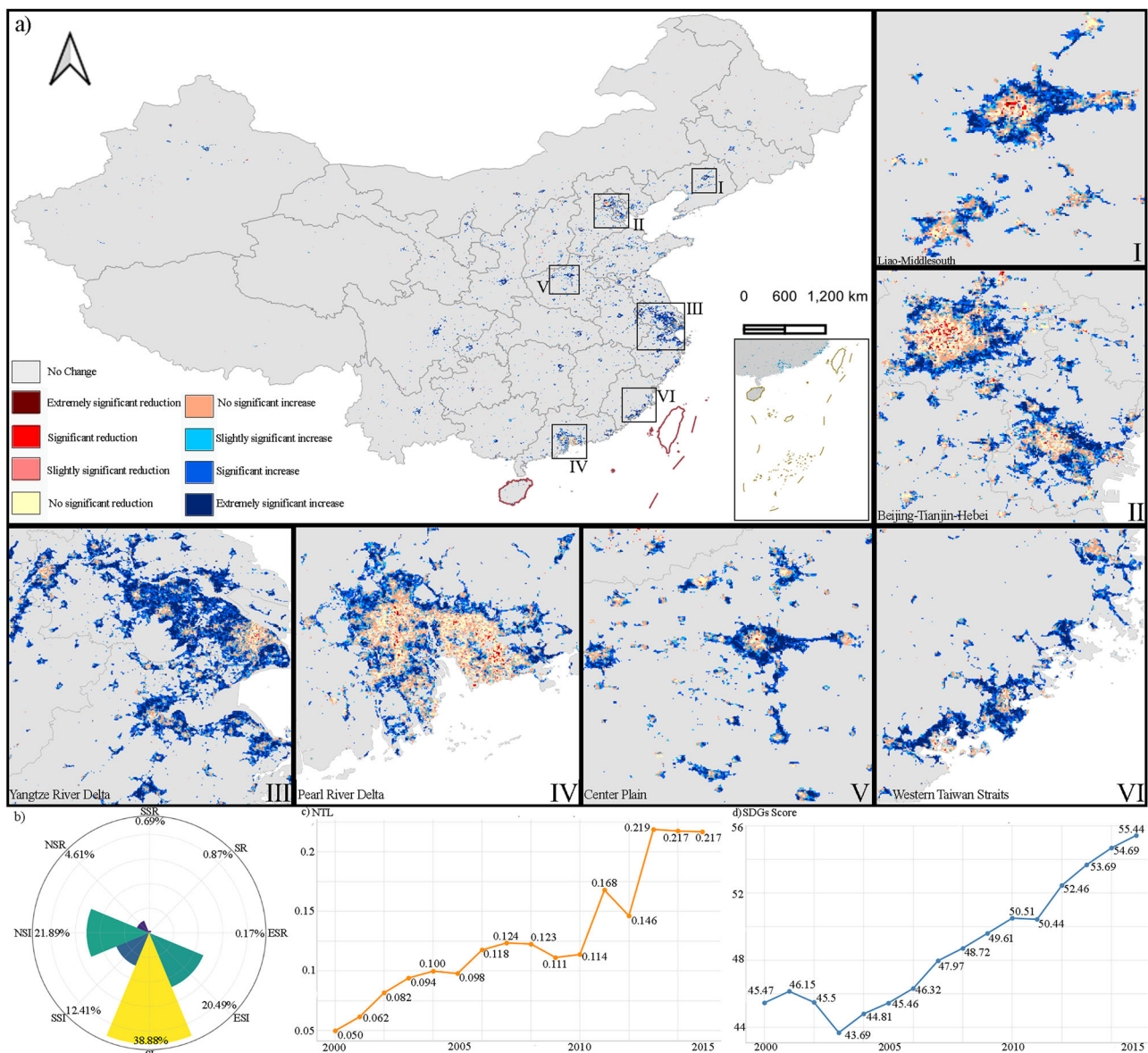


Fig. 2 NTL intensity trends across 31 subnational areas in China 2000-2015. **a** Trend analysis of NTL intensity across spatio-temporal patterns in China (Among them, I stand for the Liao-Middlesouth urban agglomeration; II stand for the Beijing-Tianjin-Hebei urban agglomeration; III stand for the Yangtze River Delta urban agglomeration; IV stand for the Pearl River Delta urban agglomeration; V stand for the Center Plain urban agglomeration; VI stand for the Western Taiwan Straits urban agglomeration). The red stand for the reduction area, and the blue stand for the increasing area. **b** Rose diagrams depict the proportion of NTL areas with different levels of change relative to the total NTL change area (the shortly name refer to the Table 1). **c** The variations in NTL average light intensity and linear fitting results showcasing values representing NTL total light intensity. **d** The variations in SDG score alongside linear fitting results, displaying average SDG score for 31 subnational area.

2010 and 2015. Consequently, synchronization between NTLI and SDG score is not evident, indicating a specific temporal lag effect (Fig. 3b, c).

The coupling and decoupling mechanism between NTLI and SDGs

Synergistic and trade-off effects of NTLI and SDGs. In the subsequent sections, we narrow our focus to the correlation between NTLI and the SDGs. In 2000, 2005, 2010, and 2015, the mean values of NTLI in China's subnational regions were 0.23, 0.26, 0.33, and 0.36, respectively (Fig. 4a). Correspondingly, the mean values of SDG score were 42.23, 45.15, 49.79, and 54.90, respectively (Fig. 4b). Regression analysis revealed a significant relationship between NTLI and SDG score ($p < 0.001$), with R^2 values

of 0.53, 0.67, 0.29, and 0.12, respectively. These results suggest that NTLI had a relatively strong explanatory power for SDG progress before 2010, whereas the relationship substantially weakened afterward. Therefore, the predictive capacity of NTLI for SDG score should be interpreted with caution (Fig. 4c).

NTLI consistently did not correlate significantly with environmental SDG score from 2000 to 2015 and lacked significance with SDG 7 (energy), SDG 10 (equity) and SDG 13 (climate) throughout the study period. Moreover, NTLI displayed an increasing correlation trend with SDG 11(city and community). NTLI showed a more substantial synergistic effect with SDG 1 (poverty), SDG 8 (economy), SDG 9 (industry), and SDG 11(city and community). This demonstrates NTLI's ability to reflect on sustainable development processes in the areas of social equity, economic development, and urban construction (Fig. 4d).

Table 4 The descriptive statistics data of NTL indices and SDG score indices in 2000, 2005, 2010, and 2020.

Indicators	Year	Mean	Median	Q1.25%	Q3.75%	Range	StD	CV
SDG score	2000	42.23	41.93	40.22	44.50	22.66	4.82	0.11
	2005	45.15	43.87	42.47	47.36	19.55	4.36	0.10
	2010	49.79	50.55	46.61	51.51	21.43	4.53	0.09
	2015	54.90	55.26	52.54	57.25	19.14	4.19	0.08
Economic SDG score	2000	29.66	27.24	25.07	31.96	31.46	7.90	0.27
	2005	34.89	32.86	29.58	38.56	33.68	7.91	0.23
	2010	41.05	38.58	36.66	44.28	31.26	7.73	0.19
	2015	46.35	44.82	41.76	50.56	24.78	6.23	0.13
Society SDG score	2000	43.41	43.20	41.24	45.66	16.80	3.67	0.08
	2005	46.82	47.06	44.46	48.93	17.82	3.82	0.08
	2010	52.67	52.44	50.62	55.50	19.94	4.25	0.08
	2015	58.38	58.24	55.15	60.83	13.94	3.57	0.06
Ecological SDG score	2000	51.30	52.14	47.58	54.05	27.06	6.20	0.12
	2005	52.03	52.97	49.36	55.10	20.02	4.74	0.09
	2010	54.59	54.50	51.29	59.20	23.66	5.56	0.10
	2015	59.03	59.90	54.85	64.77	23.00	6.96	0.12
NTLI (Nighttime Lighting Landscape Index)	2000	23.40	21.70	19.51	25.74	29.79	6.32	0.27
	2005	26.29	23.07	21.20	29.72	33.21	7.94	0.30
	2010	32.65	28.46	26.63	38.58	31.91	8.67	0.27
	2015	36.37	31.51	28.35	42.78	42.66	11.41	0.31
NTI (Nighttime Lighting Intensity Index)	2000	22.76	21.20	19.08	23.98	25.59	5.65	0.25
	2005	23.95	20.64	18.15	26.37	33.54	8.62	0.36
	2010	25.25	22.30	20.16	28.83	26.57	7.12	0.28
	2015	32.16	27.02	24.01	36.16	50.63	12.07	0.38
LI (Landscape Index)	2000	24.02	22.25	19.68	27.26	34.94	7.42	0.31
	2005	27.59	24.44	22.40	31.36	38.50	8.70	0.32
	2010	40.14	36.13	31.32	48.46	51.25	12.93	0.32
	2015	40.45	36.15	31.50	48.42	55.27	13.24	0.33

Note: The "Q1.25%" mean the 25th percentiles value, the "Q3.75%" mean the 75th percentiles value, the "Range" mean the gap between the maximum and minimum values; the "StD" mean standard deviation, and the "CV" mean coefficient of variation.

Coupling relationship between NTLI and SDGs. From 2000 to 2015, the coupled coordination degree (CCD) of NTLI with the SDG score increased from 0.54 to 0.60, and the CCD of NTLI with the economic, social, and environment dimensions of the SDG score increased from 0.47 to 0.57; 0.53 to 0.63; and 0.57 to 0.63, respectively, showing an upward trend. Geographically, developed regions such as the east coast of China predominantly displayed coupling. Since 2005, the southwest region, excluding Tibet, has progressively transitioned from a dysfunctional to a coupling relationship. However, the CCDs of subnational regions in the northwest remain low, though they show a more pronounced upward trend. High NTLI and SDG score values are concentrated in developed regions of China, such as Peking, Tientsin, Shanghai, Zhejiang, and Jiangsu (Fig. 5).

Decoupling mechanism of NTLI and SDG score. We plotted the spatial distribution of Tapio decoupling between NTLI and SDG score from 2000 to 2015. A weaker negative decoupling between NTLI and SDG score prevailed throughout this period. Both NTLI and SDG score exhibited an upward trend. However, the number of subnational regions with weak negative decoupling increased from 19 to 24, then decreased to 13, indicating that NTLI grew faster than the SDG score in 2000-2010. Additionally, the number of subnational regions with a positive coupling of NTLI and SDG score rose from 1 to 5, reflecting a gradual harmonization in their growth. Conversely, the number of weakly decoupled subnational regions declined from 7 to 3, then rose to 11, suggesting that the growth rate of SDG score gradually exceeded that of NTLI after 2010. This trend reduced the number of weakly negatively decoupled in subnational regions. As the growth rate of NTLI fell below that of SDG score, leading to a decline in their regression coefficients (Fig. 6).

Temporal trends of the CCDs. The Kernel Density Estimation (KDE) method was used to elucidate the temporal changes in CCD of NTLI and SDG Score, economic-social-environment SDG Score, respectively, from 2000 to 2015. The peak of CCD across all dimensions shifted rightward, suggesting a growing harmonization between NTLI and SDG score over time. Between 2000 and 2005, the kernel density curves for all dimensions exhibited minimal change. However, since 2010, the peaks have progressively shifted rightward at a decelerating rate, indicating that China's sustainable development began to show significant results from 2010 onwards, with increased coordination among different regions. Additionally, the kernel density curves for the society and environment dimensions have decelerated and shifted rightward since 2010, implying that social well-being and ecological protection have gradually become vital factors driving the growth of CCD between NTLI and SDG score (Fig. 7).

Spatial hot-cold areas of the CCDs. From 2000 to 2015, the distribution of cold spot and hot spot regions of the CCD of NTLI and SDG Scores, economic-social-environmental SDG score, respectively, exhibited a trend towards concentration. The hot-spots of NTLI and SDG score CCDs shifted from the northern coastal region to the middle and lower reaches of the Yangtze River, while the cold spots were primarily distributed in the northwestern and southwestern regions of China. The hotspots of NTLI and economic SDG score CCDs initially concentrated in the northern coastal region, then moved to the central region with a reduced number, and eventually further shifted southward. The distribution of cold spots and hotspots across the three dimensions of SDG score CCDs also tended to concentrate. The hot-spots of NTLI and society SDG score CCDs were predominantly located in the southwestern region, later moving from the

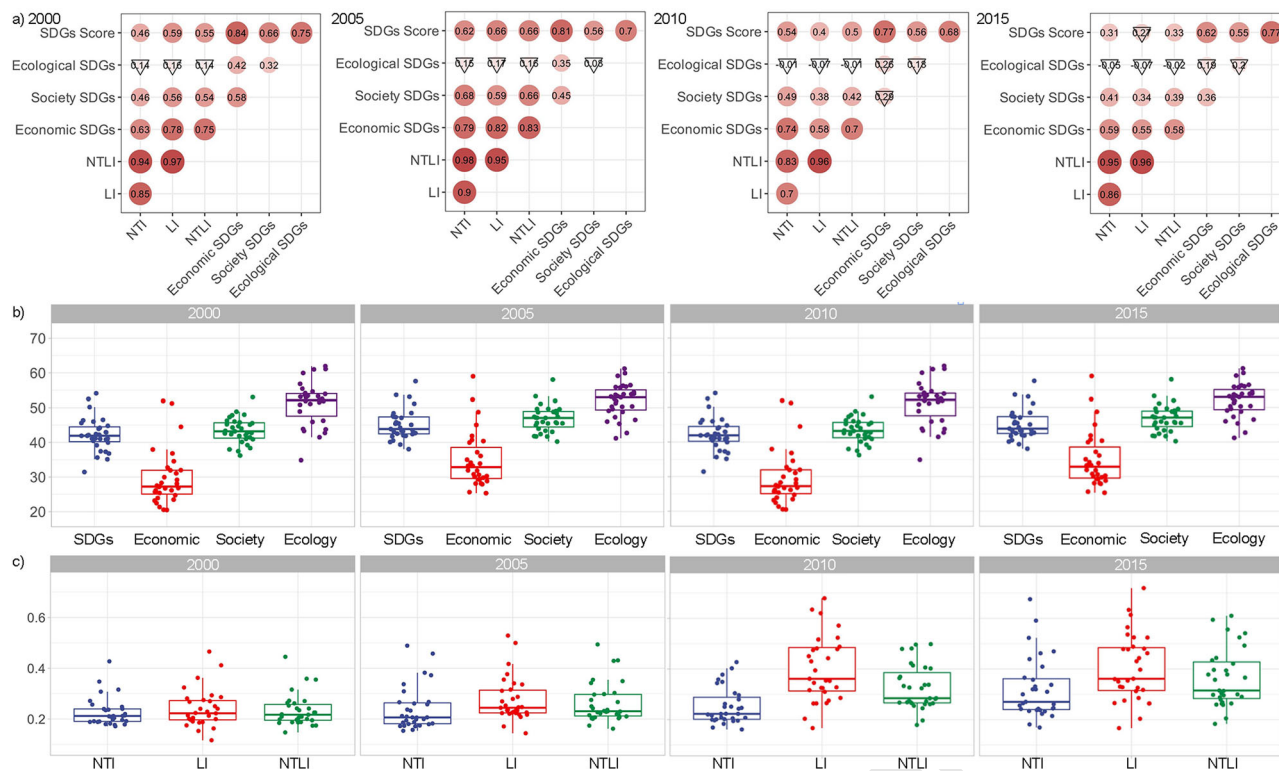


Fig. 3 The relationships between night-time light indicators and SDGs in 2000, 2005, 2010, and 2015. **a** Depiction of Spearman's correlation coefficients of NTL, LI, and NTLI with total, economic, social, and environment SDG score. Darker hues represent stronger correlations, while lighter hues denote weaker correlations, with inverted triangles indicating non-significance ($p < 0.05$). The darker the color, the stronger the correlation. **b** Illustration of box-and-line plots presenting SDG score for the economic, social, environment, and total. **c** Presentation of box plots delineating the NTL index across the NTL, LI, and NTLI dimensions.

northern coastal region to the middle and lower reaches of the Yangtze River, with cold spots expanding from the southwestern to the western region. The hotspots of CCD in NTLI and environmental SDG score shifted from the northern coastal region to the central-south and southeastern coastal regions, while cold spots remained consistently distributed in the western region of China. This evidence indicates a trend of decreasing CCD in northern China and increasing CCD in southern China, suggesting that NTLI may be more suitable for tracking SDG score in southern regions and east coastal regions (Fig. 8).

Discussion

Using the nighttime lighting data track the regional sustainable development. Compared with traditional research only consider nighttime lighting intensity (NTI) or Landscape Pattern Index (LI), Nighttime Lighting Landscape Index (NTLI) exhibits superior capacity in tracking SDG dynamics because it incorporates a more comprehensive representation of regional sustainable development by simultaneously integrating light intensity, development equity, and landscape patterns (Chen et al. 2023). The light intensity primarily captures the scale of poverty (SDG 1), energy (SDG 7), industry (SDG 8), economic (SDG 9), urbanization (SDG 11) and consumption (SDG 12). By integrating intensity-based and landscape-based information, NTLI enhances sensitivity to regional disparities and enables a more balanced assessment of development intensity and equity using raster-scale remote sensing data. This integration represents a methodological advance over prior NTL-based approaches by shifting sustainability monitoring from intensity-dominated proxies toward an equity-aware, spatially explicit framework, thereby improving the robustness and interpretability of SDG

assessments (Sun et al. 2023; Zheng et al. 2023; Noughessèwa Hounyonou 2025).

However, it is important to note that NTLI do not directly measure the normative dimensions embedded in the SDGs, such as sustainability, inclusiveness, or resilience. For example, while NTL dynamics can capture certain development-related aspects of SDG 11, such as urban growth intensity and spatial expansion, they do not fully represent all sub-goals (e.g., SDG 11.6: Reduce the adverse effects of natural disasters, SDG 11.7: Reduce the environmental impact of cities, and SDG 11.8 Provide access to safe and inclusive green and public spaces). Similarly, NTLI reflects social infrastructure-related SDGs only indirectly, by characterizing human activity intensity and spatial inequality rather than the underlying institutional, development, or social mechanisms. Additionally, NTLI reflects the spatial distribution, fragmentation, and inequality of nighttime illumination. These implications are closely linked to energy (SDG 7), equitable development (SDG 10). Therefore, NTLI captures only partial rather than comprehensive aspects of social and environmental SDGs. It should be interpreted as a spatial indicator revealing specific development-related dimensions within SDGs progress, rather than a comprehensive or direct measure of SDGs.

Synergistic and trade-off relationship between NTLI and SDG score. The continuous decline in the R^2 of NTLI and SDG score suggests the inadequacy of NTL alone in reflecting the performance levels of the 17 SDGs across economic, social, and environment dimensions (Guo et al. 2023; Yin et al. 2023; Yu et al. 2023). NTL, a remote sensing reflecting the intensity of human activities, is more suited to capturing SDGs related to economic development, such as SDG 1, SDG 9, and SDG 11 (Pan

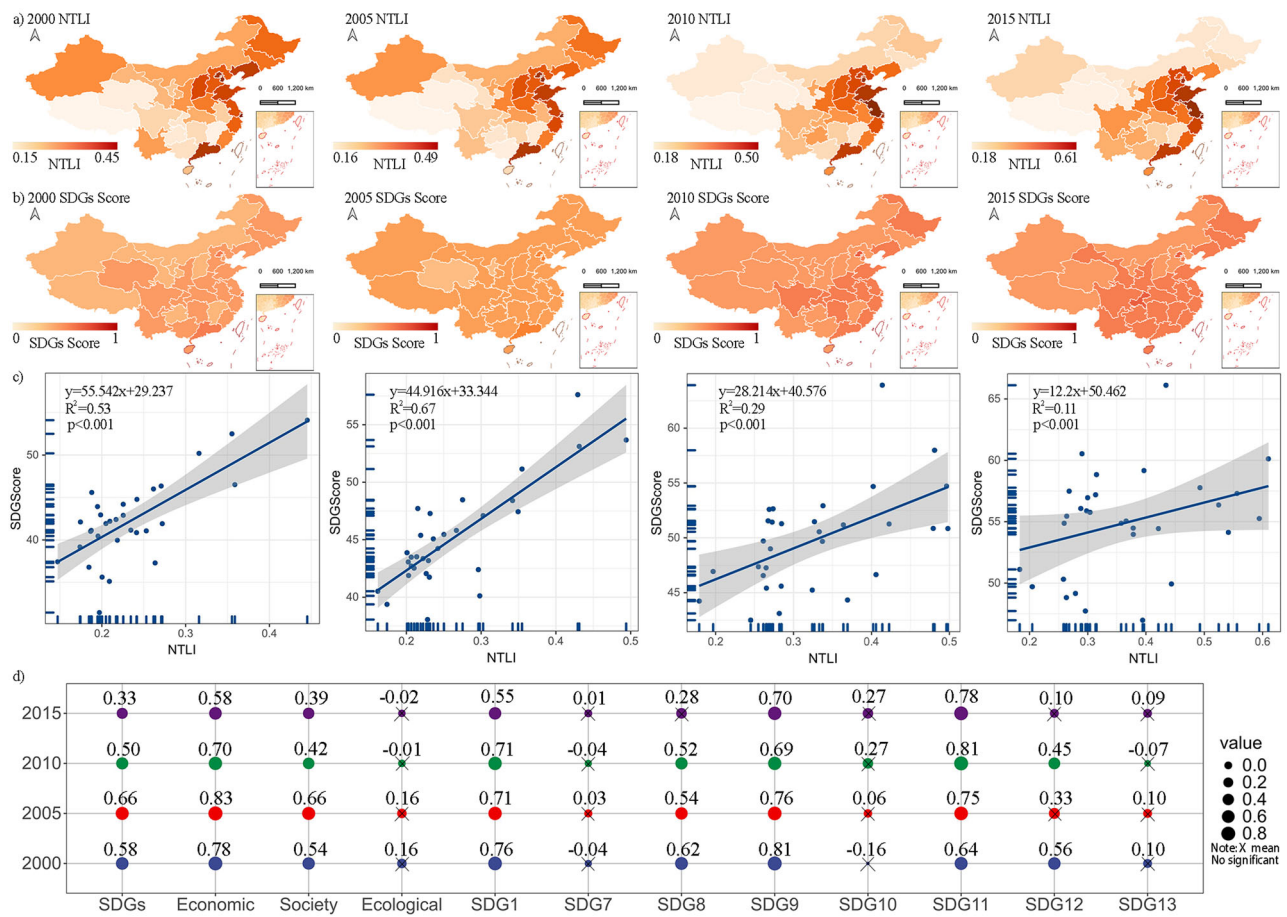


Fig. 4 Spatial-temporal changes in NTLI and SDG score and regression results for 2000, 2005, 2010, and 2015 in China's subnational regions.

a Spatial-temporal trends in NTLI. **b** Spatial-temporal trends in SDG score. **c** Linear regression results of NTLI versus SDG score. The grey intervals in the linear regression plots represent 95% confidence intervals. **d** A Spearman correlation coefficient matrix was constructed by plotting NTLI against the total, economic-social-environmental, and 8 typical SDG score. The size of the circle indicates the magnitude of Spearman's correlation coefficient, which ranges from -1 to 1. An "X" within the circle indicates no significant correlation ($p < 0.05$).

and Hu 2018; Chindarkar and Goyal 2023; Guo et al. 2023; Ferdausi et al. 2026). However, it is less effective in addressing environmental SDGs like SDG 13 (Prince 2019; Mulligan et al. 2020; Wang et al. 2024a).

Moreover, before 2010, the growth rate of NTLI outpaced that of the SDG score. Post-2010, as China's economic surpassed the Lewis inflection point, there has been a shift toward sustainable development, balancing the economic, society, and ecology. Consequently, the economic growth rate has decelerated, leading to a decline in NTLI, which better reflects the economic-social dimensions and an increase in the SDG score. This structural change slowed economic expansion and reduced NTLI growth, while SDG score continued to rise, thereby weakening the correlation between NTLI and SDG score from 2010 to 2015 and altering the coupled dynamics across regions (Tavares et al. 2021; Wang et al. 2024b; Maity and Sinha 2024).

Nevertheless, NTLI shows limited capacity in capturing environmental SDGs, such as it does not directly reflect ecosystem processes such as climate regulation, vegetation productivity, or biodiversity dynamics. We recommend leveraging the spatial-temporal consistency of remote sensing data, focusing NTL data on assessing SDGs related to economic development, social well-being, industrial growth, and urban construction, such as SDG 1, SDG 8, SDG 9, and SDG 11, rather than tracking the achievement of all SDGs. Moreover, the regression analyzes indicate that NTLI exhibited a strong

explanatory capacity for SDG score before 2010, but its predictive power declined markedly after 2010. This pattern reflect that, in the early stages, nighttime light intensity captured socio-economic growth more effectively, which strongly aligned with SDG progress. However, the relationship weakened possibly due to structural changes in growth drivers, environmental constraints, or policy interventions that decoupled with NTL and SDG score (Van Neuss 2019; Iqbal et al. 2025). We suggest that NTLI's ability to assess SDG scores should be treated with caution, and that the scope of NTLI's assessment should be narrowed down to specific SDGs.

Coupling and coordination mechanism of NTLI and SDG score. Our findings indicate a significant upward trend in the coupled coordination degree (CCD) of NTLI and SDG score over time, suggesting a strong consistency between NTLI and SDG score. The regions where CCD is rising are predominantly in southwestern China, such as Sichuan and Guizhou, with historically poor development foundations. This improvement is attributable to the Chinese government's "Western Development" policy, which has fostered economic growth and infrastructure development in these regions, thereby enhancing human well-being. Early economic development positively impacts the SDGs process, leading to a rapid increase in CCD (Lu et al. 2023). Conversely, a decline in CCD has been observed in northern China. One type of decline results from industrial transformation,

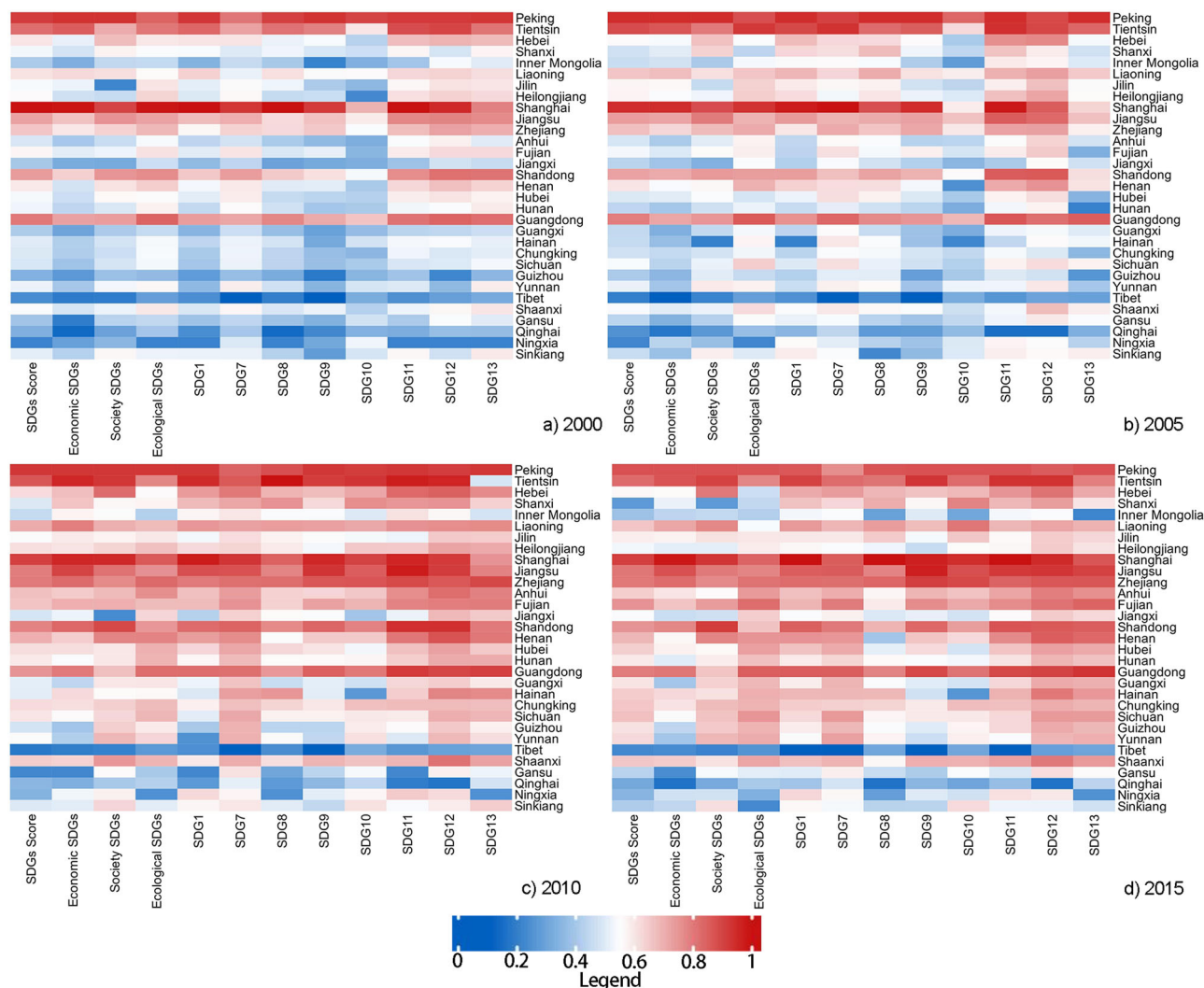


Fig. 5 CCD matrix of NTLI with the SDG score, the economic-social-environment SDG score, and 8 typical SDG scores in China 31 subnational regions. Blue indicates dysfunctional relationships and red indicates synergistic relationships, with darker shades representing more pronounced relationships. **a** CCD in 2000. **b** CCD in 2005. **c** CCD in 2010. **d** CCD in 2015.



Fig. 6 Tapio decoupling results between NTLI and SDG score, including negative decoupling, coupling, and decoupling three model to describe the trend relationship between NTLI and SDGs in 2000-2015. **a** Result in 2000-2005. **b** Result in 2005-2010. **c** Result in 2010-2015.

where the shutdown of heavy industries to improve environmental quality leads to a decrease in NTLI, as seen in regions like Hebei and Shanxi (Xiao et al. 2022). Another type stems from population exodus and economic recession, where reduced human activity results in lower NTLI, such as Heilongjiang and Jilin. The third type arises from the inequality in lighting areas due to the center agglomeration and siphon effect, causing a decrease in NTLI, notably in regions like Beijing and Guangdong (Chen et al. 2017, 2023; Tong et al. 2021; Lin et al. 2023).

The hotspot of the CCD mainly located in the eastern and southern of China exhibit a relatively high and increasingly stable

alignment between NTLI and SDG score. The observed southward shift of hotspots from the northern region toward the China East suggests that economic activity intensity and infrastructure development are more effectively translated into coordinated progress across “economic-social-environmental” SDG dimensions. Hotspot analysis enables precise identification of regions exhibiting significant clustering of high or low CCD values. High-value clusters indicate a strong alignment between NTLI and SDGs, highlighting the potential to further enhance their socio-economic synergistic effects. In contrast, coldspot clusters, characterized by low CCD values, suggest that NTLI primarily

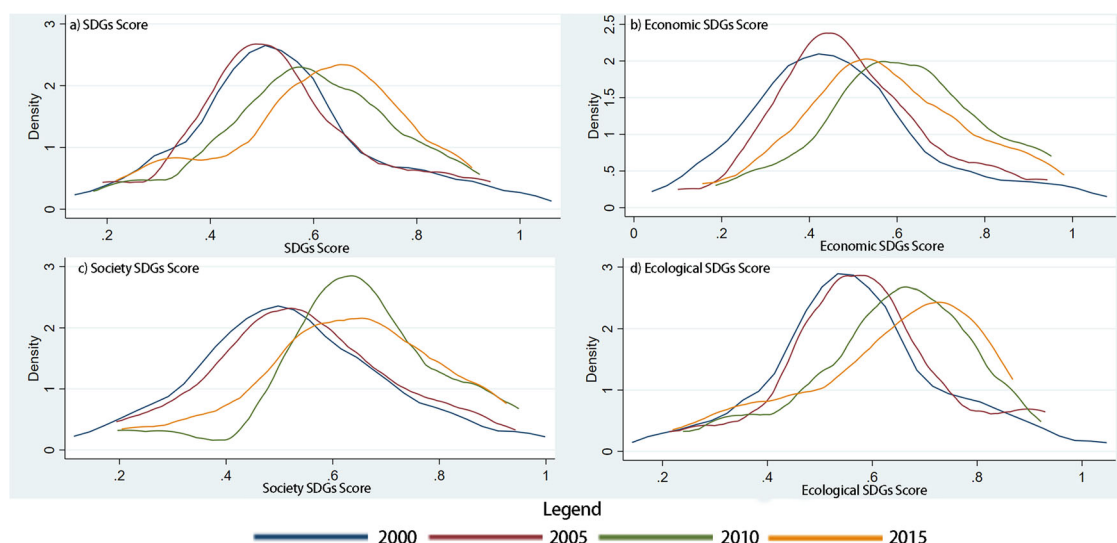


Fig. 7 Kernel density curves of China's subnational regions with CCD in 2000, 2005, 2010, and 2015 to describe the temporal trend of CCD. Among them, the blue line stands for the 2000, red line stands for the 2005, green line stands for the 2010, and orange line stand for the 2015. **a** Kernel density curves of CCD between NTLI and SDG score. **b** Kernel density curves of CCD between NTLI and economic SDG score. **c** Kernel density curves of CCD between NTLI and society SDG score. **d** Kernel density curves of CCD between NTLI and environment SDG score.

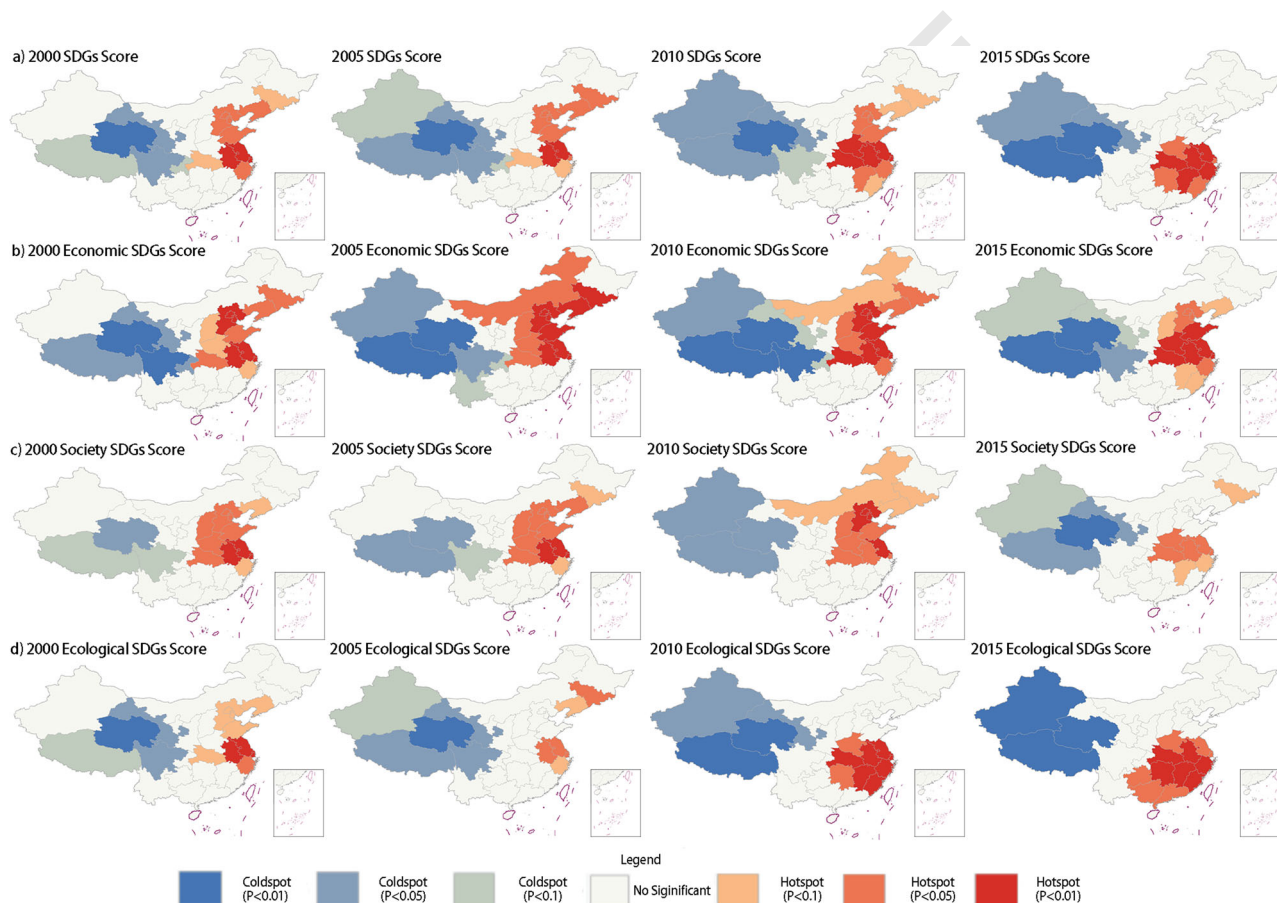


Fig. 8 Hotspot-coldspot analysis of CCD in China's subnational regions in 2000, 2005, 2010, and 2015. The blue stand for the coldspot area, and the red stand for the hotspot area. **a** Results of NTLI with SDG score's CCD. **b** The results of CCDs for NTLI with economic SDG score. **c** The results of CCDs for NTLI with society SDG score. **d** The results of CCDs for NTLI with ecological SDG score.

reflecting socioeconomic development may inadequately capture SDG progress in regions still dominated by agriculture and animal husbandry. Moreover, as NTLI is more appropriate for densely populated areas, its effectiveness in sparsely populated regions of Northwest China is limited. Therefore, careful consideration is required when using NTLI to assess SDGs in coldspot areas.

Limitations and recommendations. The primary constraint of this study arises from the utilization of NTL data, spanning 2013, a period marked by the transition from DMSP-OLS sensors to NPP-VIIRS sensors, necessitating the use of corrected production data for consistency (Levin et al. 2020). Although this dataset effectively captures the spatiotemporal variability of NTL, and was calibrated using an autoencoder to ensure a high level of consistency between the NPP-VIIRS-like data employed in this study and the original NTL observations ($R^2 = 0.87$), it does not fully encompass all information contained in the original DMSP-OLS and NPP-VIIRS datasets. Moreover, its spatial resolution of approximately 500 m limits its ability to represent fine-scale spatial details (Zheng et al. 2023). Moreover, this study devised three sets of NTL, LI, and NTLI indicators to assess the effectiveness of light intensity and landscape pattern in monitoring SDG score progression. The indicator selection aimed to identify suitable metrics for tracking SDGs progress, focusing on potential NTL intensity, development equity and landscape pattern indicators. Since the study was conducted at the raster scale, selecting too large a landscape index for Patch and Landscape could have led to bias in the NTL landscape pattern, so the study selected only the Class scale landscape pattern (Chen et al. 2017, 2023; Han et al. 2022). However, it must be stated that adding potentially suitable Patch or Landscape scale indicators may further improve the NTLI's ability to track the SDGs.

The second limitation of this study is the potential risk of partial circular reasoning. Specifically, NTLI is strongly shaped by socio-economic and human activities, such as population distribution, infrastructure, and economic development, while the composite socio-economic SDG score also incorporates several indicators derived from official statistics (McSharry and Mawejje 2024). This overlap may lead to correlations that partly reflect shared data sources rather than fully independent relationships (Yamada and Otchia 2025). Such circularity has a greater impact on the interpretation of driver factor analyses, but is less critical for the correlation and coupling-decoupling analyses, which emphasize dynamic interactions between NTLI and SDGs. Nevertheless, the possibility of circularity cannot be fully excluded. This overlap may inflate correlation strength and affect the absolute magnitude of coupling indicators, particularly in analyses focusing on driver identification and cross-sectional comparisons. The primary objective of this study is to examine the spatio-temporal co-evolution patterns and dynamic decoupling relationship between NTLI and SDGs. Nevertheless, this potential circularity represents an important methodological constraint, and future studies should further mitigate this issue by employing alternative validation datasets, excluding overlapping indicators, or applying partial correlation frameworks.

The final limitation lies in the choice of SDG score assessment instrument, wherein quantified SDG score datasets were employed to ensure data consistency and comprehensiveness (Xu et al. 2020). However, this dataset only has 2000, 2005, 2010, and 2015 four years to assure data accuracy. The absence of year-by-year China subnational detail SDG score data may have led to the oversight of crucial years or turning points from 2000 to 2015, potentially resulting in the overestimation or underestimation of NTLI's monitoring capacity for SDG score (Zhang et al. 2021;

Selin et al. 2023). Moreover, the quality of China's subnational SDG score data remains constrained by limitations in data availability, comparability, and consistency. Consequently, SDG score data beyond 2015 were not incorporated into this study, which may reduce the timeliness of the findings. Future research is therefore encouraged to integrate geospatial data with official statistical information to quantitatively characterize the near-real-time status of SDGs, enabling a more comprehensive and up-to-date assessment of subnational progress toward the 2030 Agenda.

Consequently, future investigations may consider employing data from satellites such as SDGSAT-1 and EROS-B with medium and high spatial resolutions to capture more sensitive and representative NTL details in urban areas, thereby bolstering study credibility (Guo et al. 2023; Yu et al. 2023). Furthermore, future research on NTL could include class and landscape level indicators like Shape Index (SI) and Shannon Diversity Index (SHDI) and employ statistical methods like hierarchical cluster analysis and covariance testing to refine the NTL landscape pattern index system across different landscape levels (Han et al. 2022). Moreover, researchers and organizations should aim to disclose subnational level progress performance of SDGs assessment indicators in a spatially and temporally continuous manner, utilizing remote sensing techniques like NTL as auxiliary tools for sustainability assessment to conduct unbiased quantification and analysis, thereby fostering the production of high-quality and credible SDG score products annually, enhancing fairness and transparency in sustainability research (Xu et al. 2020; Zhang et al. 2022b; Wu et al. 2022). Future research could incorporate local questionnaires or household surveys to obtain field-based data for quantifying SDG indicators. This approach would help reduce the risk of circular reasoning and enhance the accuracy and reliability of the results. Lastly, we stress that NTL should not be treated as a "magic wand". While it provides physically meaningful signals that reflect relative variations in energy consumption, economic activity, and social equality, its application scope and analytical capability remain limited and should not be overinterpreted.

Conclusions and policy recommendations

This study evaluates the effectiveness of three NTL indices (NTL, LI, and NTLI) in monitoring SDG score. It is recommended that the NTLI, which incorporates landscape pattern, development equity and light intensity, be used to track progress towards the SDGs. NTLI holds promise in assessing the SDGs process across economic and social dimensions. Findings indicate that between 2000 and 2015, 84.2% of NTL pixels in China exhibited an upward trend, notably concentrated in major urban agglomerations such as Liao-Zhong-South, Beijing-Tianjin-Hebei, the Yangtze River Delta, and the Pearl River Delta, with much of the growth occurring in urban peripheries. Moreover, while the linear correlation between NTLI and SDG score shows a declining trajectory, NTLI demonstrates heightened sensitivity to SDGs about economic development, social well-being, and urban infrastructure, such as SDG 1, SDG 8, SDG 9, and SDG 11. Furthermore, the CCD of NTLI and SDG score escalates over time, indicating spatial harmonization and a diminishing decoupling trend, reducing in inter-regional development disparities.

We recommend that the Chinese government strategically align efforts to achieve economic-social-environment dimensions sustainable development. This entails reconciling the relatively lagging environmental sector with the more rapidly advancing economic and social sectors while revising initiatives like the "Western Development Program" to bridge regional development gaps and enhance fairness. Additionally, attention should be

directed towards rejuvenating urban centers, enhancing urban infrastructure, and effectively managing population and industrial relocations. Subnational governments are urged to integrate the 169 targets and 248 indicators of the SDGs into local statistical practices to facilitate near-real-time monitoring of sustainable development progress. Finally, applying big data tools such as remote sensing, including NTL data, should be promoted in sustainability science to ensure consistent, long-term data availability, supporting ongoing monitoring of SDGs progress.

Data availability

The nighttime light data are available from the dataset at <https://doi.org/10.7910/DVN/YGIVCD>. The sustainable development goal scores are founded at <https://doi.org/10.1038/s41586-019-1846-3>. The processed research data supporting this study are included in the Supplementary Materials and deposited in the Figshare database (<https://doi.org/10.6084/m9.figshare.29994520.v1>). All analyses were conducted in the R (version 4.3.0), Fragstats (version 4.2), and ArcGIS (version 10.8).

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Competing interests

The authors declare no competing interests.

Additional information

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